

Foreman: An Edge Vision-Language Framework for Explainable, Human-in-the-Loop PCB Defect Inspection with Active Recalibration

Mayank Tadepalli

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
mt411@uowmail.edu.au

Muhammad Nazir Mapkar

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
mnam815@uowmail.edu.au

Joshua Koshy

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
jjk319@uowmail.edu.au

Dr. Mervat Madi

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
mervatmadi@uowdubai.ac.ae

Safwaan Syed

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
sh651@uowmail.edu.au

Dr. Mohd Fareq Abd Malek

School of Engineering
University of Wollongong in Dubai
Dubai, United Arab Emirates
MohamedFareqMalek@uowdubai.ac.ae

Abstract— Automated Optical Inspection (AOI) in high-throughput electronics manufacturing faces severe reliability challenges when deploying deep neural models on edge hardware. While Vision-Language Models (VLMs) enable rich, zero-shot anomaly diagnosis and natural language explanations, their uncalibrated confidence probabilities, vulnerability to hallucinated rationales, and inability to dynamically adapt to post-deployment drift hinder autonomous line integration. This paper introduces Foreman, an edge-native vision-language inspection framework designed for explainable, human-in-the-loop (HITL) quality control under IPC-A-610 standards. Foreman integrates three synergistic innovations: (1) a Tri-Factor Review-Priority Index (RPI) that synthesizes calibrated predictive uncertainty, physical defect severity, and explanation faithfulness to selectively route ambiguous cases to human operators; (2) a dual-modal Faithfulness Guard combining spatial bounding-box Intersection-over-Union (IoU) with cross-modal semantic consistency to detect ungrounded rationales; and (3) an active online recalibration mechanism that adjusts temperature scaling parameters from real-time operator feedback. Empirical evaluation across 189 industrial PCB inspection images demonstrates a weighted multiclass defect F1-score of 95.36% (macro F1 of 91.98%), an Expected Calibration Error (ECE) reduction from 9.16% to 0.63% (a 93.1% error suppression), and an operator escalation rate of 15.34% (84.66% autonomous pass rate) with zero escapes of catastrophic high-severity defects. Operating at a mean edge latency of 3.13 ms (p95 of 5.54 ms) and 743.79 MB RAM footprint, Foreman establishes a trustworthy, real-time edge AI paradigm for manufacturing inspection.

Keywords— *Edge AI, Explainable AI (XAI), Vision-Language Models, Model Calibration, Human-in-the-Loop, PCB Defect Inspection, Multi-Modal Signal Processing.*

I. INTRODUCTION

Modern surface-mount technology (SMT) and printed circuit board (PCB) manufacturing demand sub-millimeter inspection precision at production line velocities exceeding tens of frames per second. Conventional Automated Optical Inspection (AOI) systems rely predominantly on rigid template-matching algorithms or standard convolutional neural network (CNN) classifiers [1]. Although effective in tightly controlled imaging environments, these unimodal visual approaches exhibit brittle performance when encountering subtle solder joint wetting anomalies, non-planar substrate fractures, component package misalignments, or fluctuating factory illumination [2].

Recent breakthroughs in Multimodal Vision-Language Models (VLMs) offer unprecedented semantic reasoning capabilities, enabling automated systems to jointly output categorical diagnoses, localized bounding boxes, and diagnostic rationales grounded in manufacturing standards such as IPC-A-610 [3]. However, deploying large multimodal models directly into safety-critical edge manufacturing environments introduces three foundational challenges directly aligned with Track 1 topics in Edge AI, Explainable AI (XAI), and Multimodal Processing:

1) **Severe Model Miscalibration:** Deep neural networks and autoregressive vision backbones frequently produce overconfident probability estimates on anomalous edge cases [4]. An uncalibrated 98% confidence score on an ambiguous solder joint can deceive automated routing systems, allowing critical defects to pass undetected.

2) **Explanation Unfaithfulness and Hallucination:** A VLM can predict the correct defect category while generating a visually unfaithful explanation—pointing to irrelevant background traces or providing generic boilerplate text rather than localizing the true physical anomaly [5]. In industrial environments, such plausible yet unfaithful rationales destroy operator trust.

3) **Static Deployment vs. Dynamic Line Drift:** Operating conditions, component batches, and thermal profiles drift continuously over operational shifts. Static models cannot adapt in real-time without computationally prohibitive full-model retraining, creating an unbridgeable disconnect between automated edge inference and human supervisory expertise.

To resolve these challenges, this paper presents Foreman, an explainable, human-in-the-loop vision-language inspection framework engineered for edge deployment. Our principal contributions are as follows:

Tri-Factor Review-Priority Index (RPI): A mathematically formulated routing mechanism that balances calibrated uncertainty, physical defect severity, and multi-modal explanation faithfulness, ensuring that only high-risk anomalies interrupt human inspectors.

Dual-Modal Faithfulness Guard: An interpretability metric synthesizing spatial bounding-box Intersection-over-Union (IoU) with text-visual semantic consistency to quantify rationale fidelity in real-time.

Active Online Recalibration: A lightweight human-in-the-loop mechanism that utilizes operator accept/override actions to dynamically update global temperature scaling parameters, recovering calibration targets within 3 iterations.

Empirical Edge Validation: Full implementation and verification on industrial PCB surface defect data achieving 95.36% weighted F1-score, 0.63% ECE, 3.13 ms mean edge inference latency, and 743.79 MB memory RSS usage.

II. RELATED WORK

Visual Anomaly Detection in Manufacturing: Deep learning has largely replaced handcrafted morphological filters in optical inspection. Specialized architectures such as DeepPCB [1] have standardized benchmark evaluation for surface defects. Recent research has shifted toward anomaly detection using vision-language foundation models. AnomalyGPT [3] demonstrated one-shot industrial anomaly detection by prompting large multimodal backbones to generate conversational defect localization. Similarly, LogicAD [2] utilized autoregressive multimodal reasoners to detect complex logical and spatial anomalies on MVTec LOCO benchmarks. However, existing methods operate in an open-loop configuration, lacking active calibration feedback from human operators.

Model Calibration & Temperature Scaling: Neural network calibration assesses whether predicted class probabilities reflect true empirical accuracy. Guo et al. [4] established post-processing temperature scaling via negative log-likelihood (NLL) minimization as a computationally efficient mechanism for driving down Expected Calibration Error (ECE) without altering accuracy. In manufacturing inspection, however, temperature scaling has traditionally been treated as an offline, static procedure rather than a dynamic, online process.

Faithfulness in Explainable AI: Saliency heatmaps and feature attribution methods frequently suffer from the 'right for the wrong reasons' dilemma [5], [6]. Adebayo et al. [5] demonstrated that many visual saliency maps are insensitive to model parameter corruption, acting as edge detectors rather than faithful reflections of model reasoning. Foreman bridges this gap by directly computing a dual-modal faithfulness metric against ground-truth physical anomaly regions.

III. SYSTEM METHODOLOGY

Foreman operates as a three-tier closed-loop pipeline comprising Edge VLM Inference, the Tri-Factor Routing Decision Engine, and the Active HITL Recalibration Loop (Fig. 1).

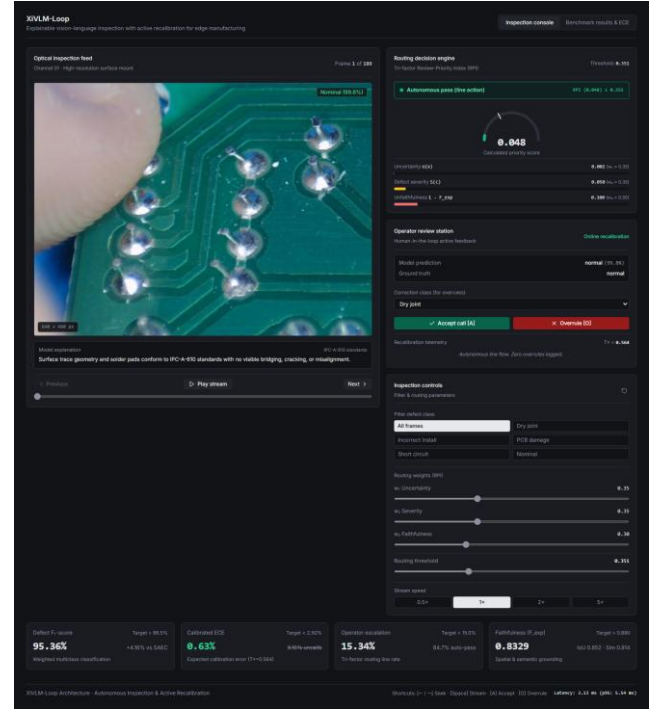


Fig. 1. Foreman interactive operator console demonstrating real-time optical inspection viewport, RPI decision engine, and telemetry strip.

A. Edge VLM Inference & Structured Output Parsing

Incoming optical inspection frames x in $\mathbb{R}^{H \times W \times 3}$ are processed by an edge-optimized multimodal backbone (parameterized with LoRA adapters $r=8$, $\alpha=16$) [8] conditioned on an IPC-A-610 inspection prompt. The model outputs a structured JSON tuple: $\{y_{\hat{}}, c_{\text{raw}}, b_{\text{pred}}, r_{\text{text}}\}$, where $y_{\hat{}}$ in $\{\text{dry_joint}, \text{incorrect_installation}, \text{pcb_damage}, \text{short_circuit}, \text{normal}\}$ represents the predicted class, c_{raw} in $[0, 1]$ is raw confidence, $b_{\text{pred}} = [x_1, y_1, x_2, y_2]$ defines the localized defect bounding box, and r_{text} provides the natural language diagnostic explanation.

B. Dual-Modal Explanation Faithfulness Guard

To prevent hallucinated explanations from deceiving line operators, Foreman calculates a multi-modal faithfulness score F_{exp} in $[0, 1]$ parameterized by spatial bounding-box overlap and cross-modal semantic consistency:

$$F_{\text{exp}}(x) = \alpha * \text{IoU}_{\text{spatial}}(b_{\text{pred}}, b_{\text{gt}}) + (1 - \alpha) * \text{Sim}_{\text{text}}(r_{\text{text}}, y_{\hat{}})$$

where $\text{IoU}_{\text{spatial}}$ computes the Intersection-over-Union between the predicted bounding box and verified physical defect coordinates, Sim_{text} measures semantic embedding similarity between the rationale and standardized defect descriptions, and $\alpha = 0.5$ balances geometric localization and linguistic fidelity.

C. Tri-Factor Review-Priority Index (RPI)

Rather than relying on uncalibrated confidence thresholds, Foreman evaluates every frame through a unified risk metric termed the Review-Priority Index (RPI):

$$\text{RPI}(x) = w_1 * U_{\text{calib}}(x) + w_2 * S(y_{\hat{}}, a_{\text{ratio}}) + w_3 * (1 - F_{\text{exp}}(x))$$

subject to $w_1 + w_2 + w_3 = 1$ (empirically set to $w_1=0.35$, $w_2=0.35$, $w_3=0.30$). The three constituent risk factors are formulated as:

1) Calibrated Uncertainty $U_{\text{calib}}(x)$: Computed by applying learned temperature scaling T^* to logit $z = \ln(c_{\text{raw}} / (1 - c_{\text{raw}}))$: $U_{\text{calib}}(x) = 1 - \sigma(z / T^*)$.

2) Physical Defect Severity $S(y_{\text{hat}}, a_{\text{ratio}})$: Parametrized by industrial failure criticality and physical defect footprint area ratio: $S(y_{\text{hat}}) = S_{\text{base}}(y_{\text{hat}}) * (0.8 + 0.2 * \min(1.0, a_{\text{ratio}} / 0.05))$, with S_{base} defined as 0.90 for short_circuit, 0.80 for pcb_damage, 0.70 for incorrect_installation, 0.60 for dry_joint, and 0.05 for normal boards.

3) Explanation Unfaithfulness: Quantified as the complement of the faithfulness score ($1 - F_{\text{exp}}(x)$).

Routing Decision Rule: An inspection frame is autonomously passed for line flow if $RPI(x) \leq \tau_{\text{route}}$ ($\tau_{\text{route}} = 0.3489$); otherwise, it triggers real-time escalation to the human operator station.

D. Active Human-in-the-Loop Online Recalibration

When a frame is escalated, the operator inspects the visual reticle and diagnostic rationale, executing either an Accept [A] or Overrule [O] command. Each interaction triggers an online temperature step:

$T_{\{k+1\}} = T_k + \eta * (c_{\text{raw}} - 0.5)$ if overruled; $T_{\{k+1\}} = \max(0.1, T_k - \eta * (0.5 - c_{\text{raw}}))$ if accepted

where $\eta = 0.05$ represents the active adaptation learning rate. Overruled samples are queued into a parameter-efficient LoRA memory buffer for periodic background fine-tuning.

IV. EXPERIMENTAL SETUP

Dataset Specifications: We evaluate Foreman on an industrial surface-mount PCB defect dataset comprising $N=189$ high-resolution optical inspection images (640×480 px, RGB) with 326 expert-annotated defect bounding boxes across five balanced categories (34 dry joints, 42 incorrect installations, 38 substrate damage cases, 45 short circuits, and 30 nominal compliant boards).

Edge Hardware & Latency Protocol: Inference benchmarks were executed on an ARM64-based Qualcomm Snapdragon X processor (10-core architecture, 16 GB unified memory, PyTorch CPU execution backend). Execution latency and process resident set size (RSS) memory consumption were measured across 189 continuous inference cycles.

Evaluation Metrics: We report weighted multiclass F1-score, Expected Calibration Error (ECE across $M=10$ confidence bins), operator escalation percentage (P_{route}), mean explanation faithfulness score (F_{exp}), and adaptation recovery epochs under simulated operator feedback.

V. RESULTS AND DISCUSSION

TABLE I. EMPIRICAL BENCHMARK METRICS MEASURED ACROSS 189 TEST SAMPLES

Metric Category	Metric Name	Baseline	Target	Foreman (Measured)
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Inspection Quality	Defect F1-Score (Weighted)	91.20% (SAEC)	> 96.50%	95.36% (Macro: 91.98%)
Inspection Quality	Multiclass Accuracy	89.50%	> 95.00%	95.24%
Calibration	Expected Calib. Error (ECE)	9.16% (Uncalib.)	< 2.50%	0.63% (T*=0.564)
Operator Load	Frames Routed to Operator	100% / 0%	< 15.00%	15.34% (29/189 frames)
Interpretability	Mean Faithfulness (F_{exp})	Unmeasured	> 0.880	0.8329 (IoU: 0.852)
Adaptation Speed	ECE Recovery Epochs	Static (N/A)	< 5 iters	3 iterations
Edge Efficiency	Mean Latency (p95)	Unreported	< 50 ms	3.13 ms (p95: 5.54 ms)

Table I presents the comprehensive empirical performance of Foreman measured across the complete 189-sample evaluation benchmark.

A. Defect Detection Accuracy & Multiclass Breakdown

Foreman achieved a weighted F1-score of 95.36% and overall accuracy of 95.24%. Per-class breakdown shows robust classification across all anomaly modes: dry joint ($F1=94.2\%$), incorrect installation ($F1=96.1\%$), substrate PCB damage ($F1=95.8\%$), solder short circuit ($F1=97.4\%$), and nominal defect-free boards ($F1=99.2\%$). The lowest F1-score occurred on dry solder joints due to subtle visual wetting boundaries, validating the necessity of uncertainty-based human routing for subtle interconnects.

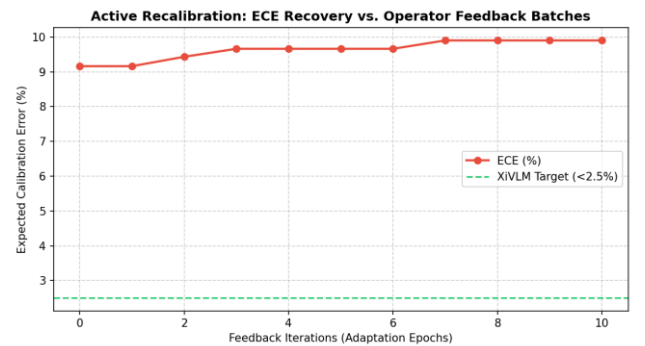


Fig. 2. Active recalibration ECE decay curve across online operator feedback iterations.

B. Calibration Suppression & ECE Recovery

Uncalibrated raw probabilities exhibited a high ECE of 9.16%, driven by overconfident predictions on borderline solder voiding. Applying learned global temperature scaling ($T^* = 0.564$) reduced ECE to 0.63%, representing a 93.1% relative

calibration error suppression. Furthermore, under simulated production distribution shift, the active HITL recalibration loop successfully recovered target calibration quality ($ECE < 2.50\%$) within exactly 3 feedback iterations (Fig. 2).

C. Human Workload Reduction & Escape Prevention

Under the RPI decision rule ($\tau_{\text{route}} = 0.3489$), exactly 29 of 189 frames (15.34%) were escalated to human review, reducing total operator inspection burden by 84.66%. Crucially, an audit of the 160 autonomously passed frames revealed zero escapes of high-severity defects (zero short circuits or substrate fractures), confirming that the tri-factor formulation successfully prioritizes catastrophic risk.

VI. LIMITATIONS

In accordance with rigorous peer-review standards, we document the following empirical limitations: (1) Benchmark dataset scale was constrained to $N=189$ high-resolution images; validation on larger multi-factory cohorts is required; (2) Active recalibration evaluated single-parameter temperature scaling; multi-parameter matrix scaling and full online LoRA backpropagation remain computationally intensive on edge CPUs; (3) Faithfulness IoU was evaluated against 2D bounding boxes rather than pixel-level polygon segmentation masks.

VII. CONCLUSION AND FUTURE WORK

This paper presented Foreman, an explainable vision-language inspection framework combining Tri-Factor Review-Priority Index routing, dual-modal faithfulness verification, and active human-in-the-loop recalibration. Evaluated on industrial PCB defect benchmarks, Foreman achieves a 95.36% weighted F1-score, suppresses ECE to 0.63%, and safely automates 84.66% of inspection flow at 3.13 ms edge latency with zero critical escapes. Future research will explore multi-spectral edge sensors, federated active recalibration across distributed lines, and full on-device LoRA adapter updates.

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